

$$1) \quad a) \quad \lim_{x \rightarrow \infty} 1/x \cdot \cos(1/x) : \quad \lim_{x \rightarrow \infty} (1/x) = 0 \wedge \lim_{x \rightarrow 0} \cos(x) = 1$$

$$\Rightarrow \lim_{x \rightarrow \infty} 1/x \cdot \cos(1/x) = 0 \cdot 1 = 0$$

$$b) \quad \lim_{x \rightarrow \infty} 1/x \cdot \cos(x) : \quad \lim_{x \rightarrow \infty} (1/x) = 0 \wedge \lim_{x \rightarrow \infty} \cos(x) \in [-1; 1]$$

$$\Rightarrow \lim_{x \rightarrow \infty} 1/x \cdot \cos(x) = 0 \cdot [-1; 1] = 0$$

$$c) \quad \lim_{x \rightarrow 0} x \cdot e^x : \quad \lim_{x \rightarrow 0} x = 0 \wedge \lim_{x \rightarrow 0} e^x = 1$$

$$\Rightarrow \lim_{x \rightarrow 0} x \cdot e^x = 0 \cdot 1 = 0$$

$$d) \quad \lim_{x \rightarrow -\infty} x \cdot e^x : \quad \lim_{x \rightarrow -\infty} x = -\infty \wedge \lim_{x \rightarrow -\infty} e^x = 0$$

$$\Rightarrow \lim_{x \rightarrow -\infty} x \cdot e^x = -\infty \cdot 0 = 0$$

$\downarrow$ $\downarrow$
 linear $\leftrightarrow$ exponential (Dominanzprinzip)

$$2) a) \lim_{h \rightarrow \infty} \frac{\cos(h)}{L_h(h)} : \lim_{h \rightarrow \infty} \cos(h) \in [-1; 1] \wedge \lim_{h \rightarrow \infty} L_h(h) = \infty$$

$$\Rightarrow \lim_{h \rightarrow \infty} \frac{\cos(h)}{L_h(h)} = \frac{[-1; 1]}{\infty} = 0$$

$$b) \lim_{h \rightarrow \infty} \sin(1/h) \cdot \cos(h) : \lim_{h \rightarrow \infty} \cos(h) \in [-1; 1] \wedge \lim_{h \rightarrow \infty} \sin(1/h) = \sin(0) = 0$$

$$\Rightarrow \lim_{h \rightarrow \infty} \sin(1/h) \cdot \cos(h) = 0 \cdot [-1; 1] = 0 \quad \left(\begin{array}{l} \text{beschränkte Folge mal Nullfolge} \\ \text{ergibt Nullfolge} \end{array} \right)$$

$$c) \lim_{h \rightarrow \infty} \frac{h^4 - 3h^2}{h^3 - 3h^4} \cdot e^{\frac{2}{h^3+1}} : \left. \begin{array}{l} \lim_{h \rightarrow \infty} \frac{h^4 - 3h^2}{h^3 - 3h^4} = \lim_{h \rightarrow \infty} \frac{h^4 \cdot (1 - 3/h^2)}{h^4 \cdot (1/h - 3)} = -1/3 \\ \lim_{h \rightarrow \infty} e^{\frac{2}{h^3+1}} = e^{0/60} = e^0 = 1 \end{array} \right\} \Rightarrow -1/3 \cdot 1 = -1/3$$

$$d) \lim_{h \rightarrow \infty} h \cdot L_h(1 + 1/h) = \lim_{h \rightarrow \infty} L_h(1 + 1/h)^h$$

$$\lim_{h \rightarrow \infty} (1 + 1/h)^h = e^1, \text{ da } \lim_{h \rightarrow \infty} \left(1 + \frac{x}{h}\right)^h = e^x$$

$$\Rightarrow \lim_{h \rightarrow \infty} h \cdot L_h(1 + 1/h) = L_h e^1 = 1$$

$$3) a) f(x) = a^x = (e^{\ln a})^x = e^{\ln a \cdot x} \quad ; x \in \mathbb{R} ; a > 0$$

$f(x)$ ist stetig, da jedes ganzrationales Polynom und jede e-Funktion stetig ist.

$$b) \lim_{k \rightarrow \infty} \sqrt[k]{k} = 1, \text{ da } \sqrt[k]{k} = k^{1/k} = (e^{\ln k})^{1/k} = e^{\overbrace{\ln k}^{\infty} \cdot \overbrace{1/k}^0} = e^0 = 1$$

$$4) a) a^x \cdot a^y = (e^{\ln a})^x \cdot (e^{\ln a})^y = e^{\ln a \cdot x} \cdot e^{\ln a \cdot y} = e^{\ln a \cdot x + \ln a \cdot y} = e^{\ln a \cdot (x+y)} \\ = (e^{\ln a})^{x+y} = e^{x+y}$$

$$b) (a^x)^y = (e^{\ln a^x})^y = e^{y \cdot \ln a^x} = e^{y \cdot \ln (e^{\ln a})^x} = e^{y \cdot \ln e^{x \cdot \ln a}} = e^{y \cdot x \cdot \ln a} = e^{x \cdot y \cdot \ln a} = a^{x \cdot y}$$

$$c) a^x \cdot b^x = e^{x \cdot \ln a} \cdot e^{x \cdot \ln b} = e^{x \cdot \ln a + x \cdot \ln b} = e^{x \cdot (\ln a + \ln b)} = e^{x \cdot \ln (a \cdot b)} = (a \cdot b)^x$$

$$5) a) (i) \lg_{10}(10) = 1 \quad (ii) \lg_{10}(10.000) = \lg_{10}(10^4) = 4$$

$$(iii) \lg_{10}(1) = \lg_{10}(10^0) = 0 \quad (iv) \lg_{10}(0,01) = \lg_{10}(10^{-2}) = -2$$

$$b) (i) \lg_2(1/64) = \lg_2(1/32) = \lg_2(2^{-5}) = -5 \quad (ii) \lg_4(2/64) = \lg_4(2) - \lg_4(64) = \lg_4 4^{1/2} - \lg_4 4^3 = 1/2 - 3 = -5/2$$

$$(iii) \lg_{32}(2/64) = \lg_{32}(1/32) = \lg_{32}(32)^{-1} = -1 \quad (iv) \lg_{64}(2/64) = \lg_{64} 2 - \lg_{64} 64 = \lg_{64} 64^{1/6} - \lg_{64} 64^1 = 1/6 - 1 = -5/6$$

$$5) \text{ c) i) } \log_2 64 = \log_2 2^6 = 6$$

$$\text{ii) } \frac{\log_2(2/64)}{\log_2(64)} = \frac{\log_2 2^{-5}}{\log_2 2^6} = -\frac{5}{6} = \log_{64} (2/64) \quad \Rightarrow \quad \frac{\log_a x}{\log_a b} = \log_b x$$

$$6) \text{ a) } \sin(x) = \sum_{k=0}^{\infty} (-1)^k \cdot \frac{x^{2k+1}}{(2k+1)!}$$

$$\left| \frac{a_{k+1}}{a_k} \right| = \left| \frac{(-1)^{k+1} \cdot x^{2(k+1)+1} \cdot (2k+1)!}{(2 \cdot (k+1) + 1)! \cdot (-1)^k \cdot x^{2k+1}} \right| = \left| \frac{(-1)^{k+1}}{(-1)^k} \cdot \frac{x^{2k+3}}{x^{2k+1}} \cdot \frac{(2k+1)!}{(2k+3)(2k+2)(2k+1)!} \right|$$

$$\Rightarrow \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = \lim_{k \rightarrow \infty} \frac{x^2}{(2k+3)(2k+2)} = 0 < 1$$

$$\cos(x) = \sum_{k=0}^{\infty} (-1)^k \cdot \frac{x^{2k}}{(2k)!}$$

$$\left| \frac{a_{k+1}}{a_k} \right| = \left| \frac{(-1)^{k+1} \cdot x^{2(k+1)} \cdot (2k)!}{(2 \cdot (k+1))! \cdot (-1)^k \cdot x^{2k}} \right| = \left| \frac{(-1)^{k+1}}{(-1)^k} \cdot \frac{x^{2k+2}}{x^{2k}} \cdot \frac{(2k)!}{(2k+2)(2k+1)(2k)!} \right|$$

$$\Rightarrow \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = \lim_{k \rightarrow \infty} \frac{x^2}{(2k+2)(2k+1)} = 0 < 1$$

b)

$$\cos(-x) = \sum_{k=0}^{\infty} (-1)^k \cdot \frac{(-x)^{2k}}{(2k)!} = \sum_{k=0}^{\infty} (-1)^k \cdot \frac{(-1)^{2k} \cdot (x)^{2k}}{(2k)!} = \sum_{k=0}^{\infty} (-1)^k \cdot \frac{(x)^{2k}}{(2k)!} = \cos(x)$$

$\Rightarrow$ Achsensymmetrie $f(x) = f(-x)$

$$\sin(-x) = \sum_{k=0}^{\infty} (-1)^k \cdot \frac{(-x)^{2k+1}}{(2k+1)!} = \sum_{k=0}^{\infty} (-1)^k \cdot \frac{(-1)^{2k+1} \cdot (x)^{2k+1}}{(2k+1)!} = (-1) \cdot \sum_{k=0}^{\infty} (-1)^k \cdot \frac{(x)^{2k+1}}{(2k+1)!} = -\sin(x)$$

$\Rightarrow$ Punktsymmetrie $f(-x) = -f(x)$